

Project 2– Design and Deployment of a Highly Available and Scalable Web Application on AWS

Prepared By: Kritika Agarwal

Project: Design and Deployment of a Highly Available and Scalable Web Application on AWS.

Objective

- Login AWS
- Project Purpose

Architecture Structure

- Structure
- Explanation

AWS Services Used

- Amazon EC2
- Elastic Load Balancer (Application Load Balancer)
- Auto Scaling Group
- Amazon VPC
- Amazon RDS
- Amazon S3
- Amazon CloudWatch
- Amazon SNS
- AWS IAM
- Amazon Route 53

Testing Outcome

- Uploaded
- Created

Conclusion

- Summary
- Learning

1.OBJECTIVE-

- Login to AWS Console
- Search EC2
- Open the EC2 Dashboard.

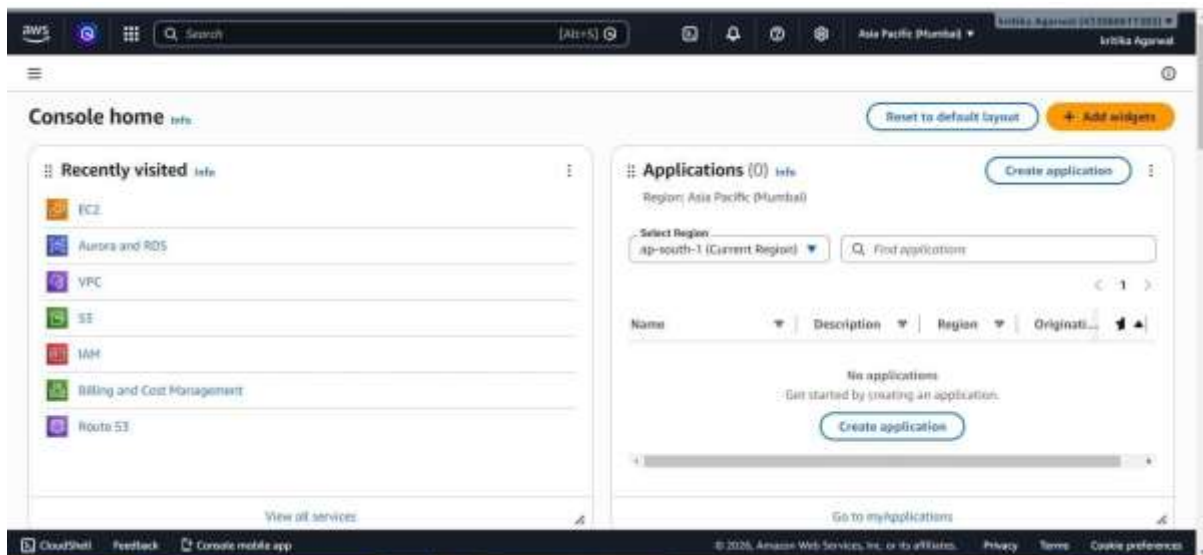


FIG-1: AWS CONSOLE HOME PAGE

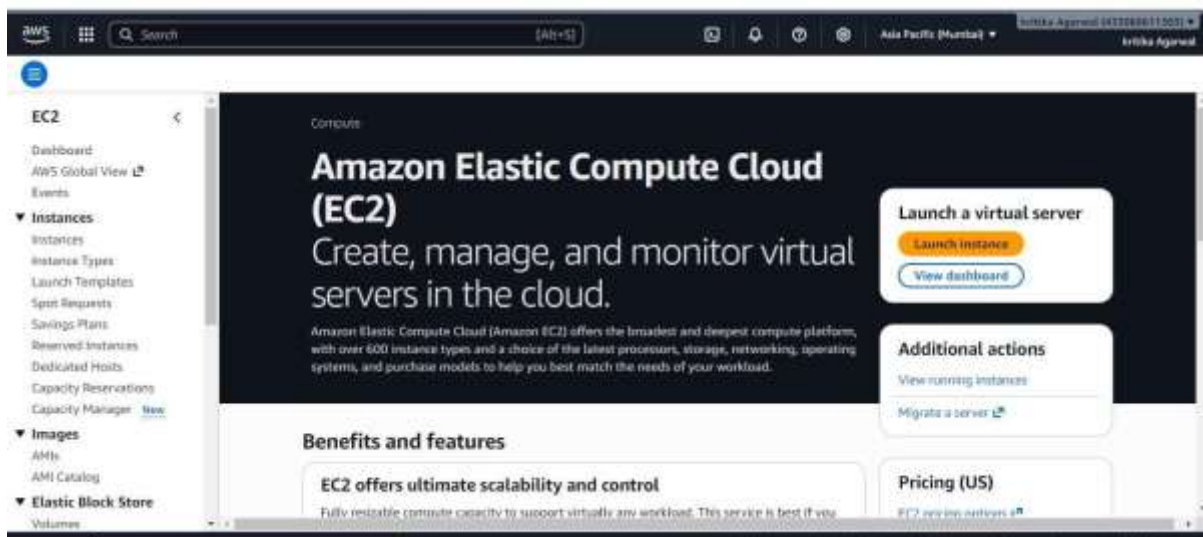
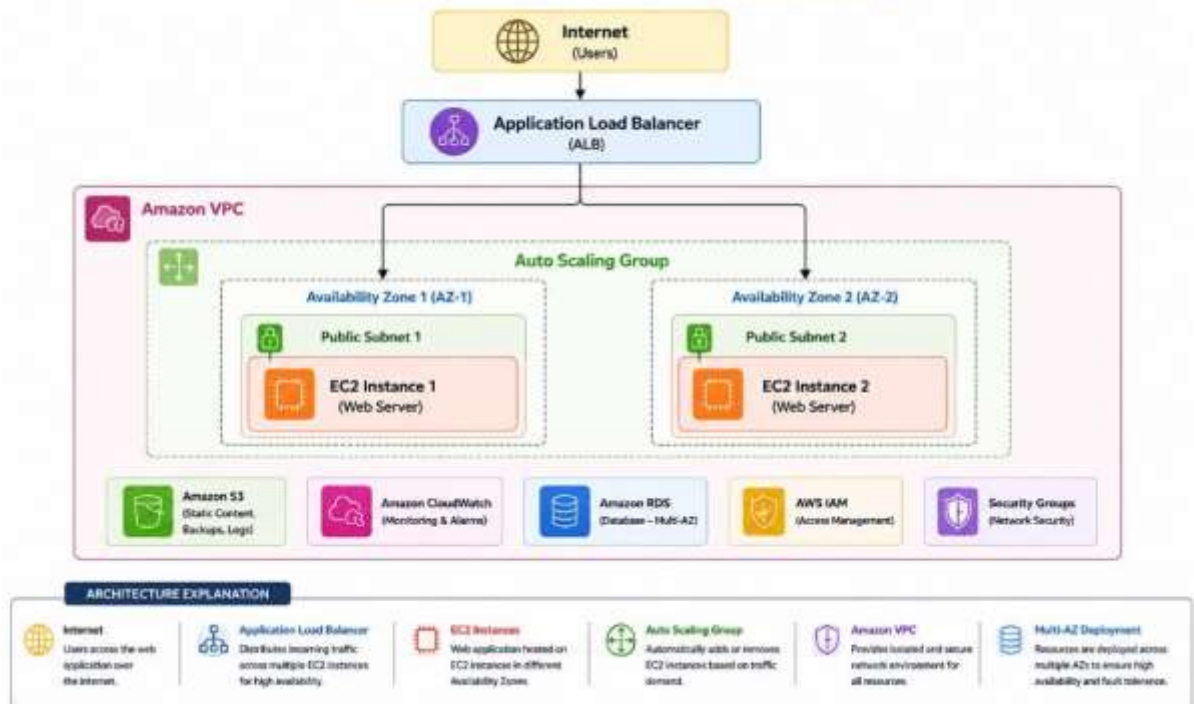


FIG-2: EC2 DASHBOARD

2. ARCHITECTURE STRUCTURE:

- To design a production-grade cloud architecture.
- To deploy a web application on AWS.
- To ensure High Availability of the application
- To implement Auto Scaling for handling different workloads.
- To secure the infrastructure using Amazon VPC and Security Groups.

Project 2 – Design and Deployment of a Highly Available and Scalable Web Application on AWS



3. AWS SERVICES USED:

Step 1: Create Amazon VPC-

- Create a new VPC.
- Create Public and Private Subnets.
- Attach an Internet Gateway.
- Configure Route Tables.
- Configure Security Group.

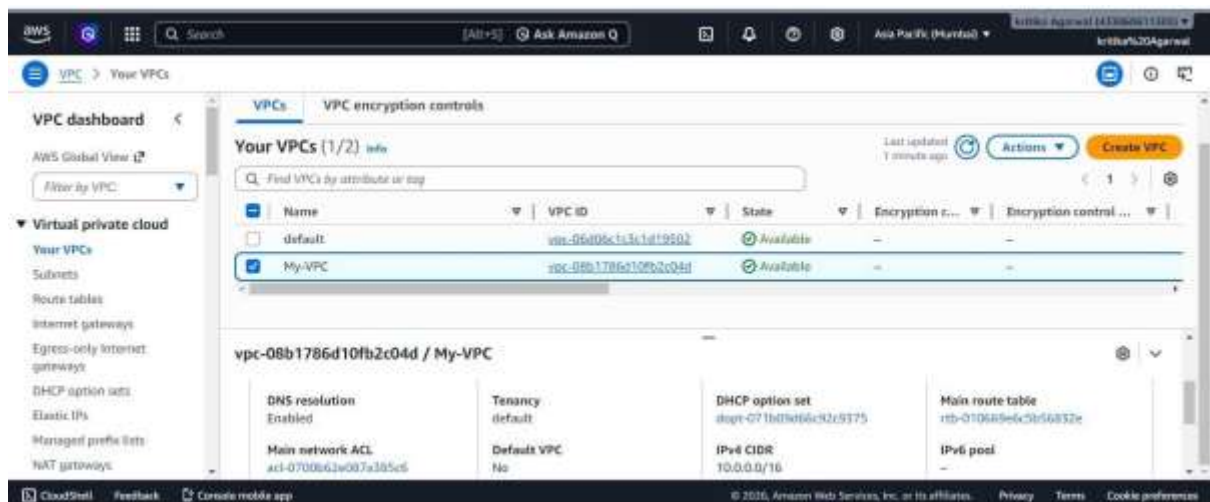


FIG-1: VPC

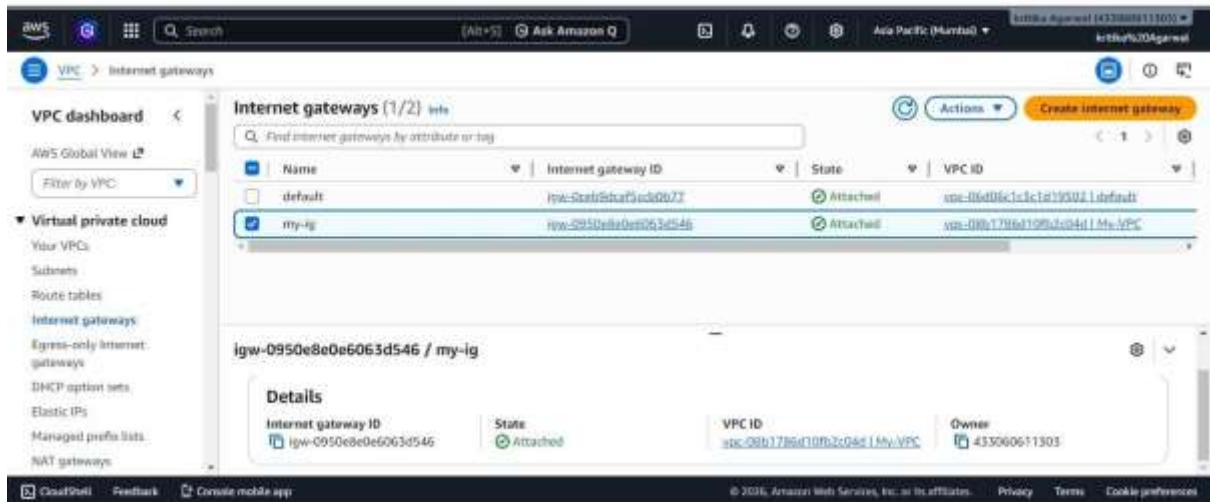


FIG-2: IGW

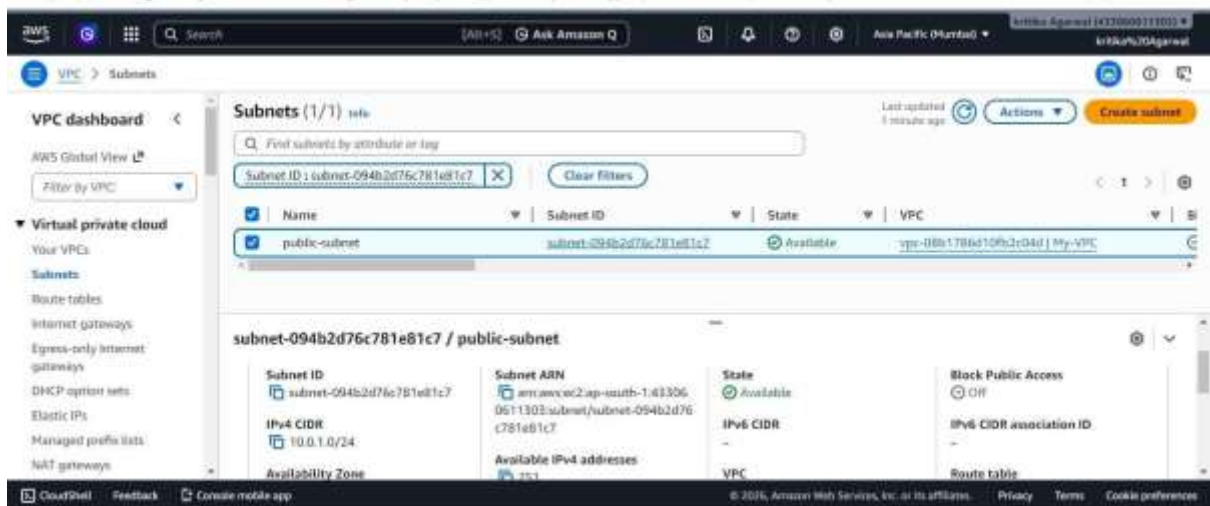


FIG-3: PUBLIC-SUBNET

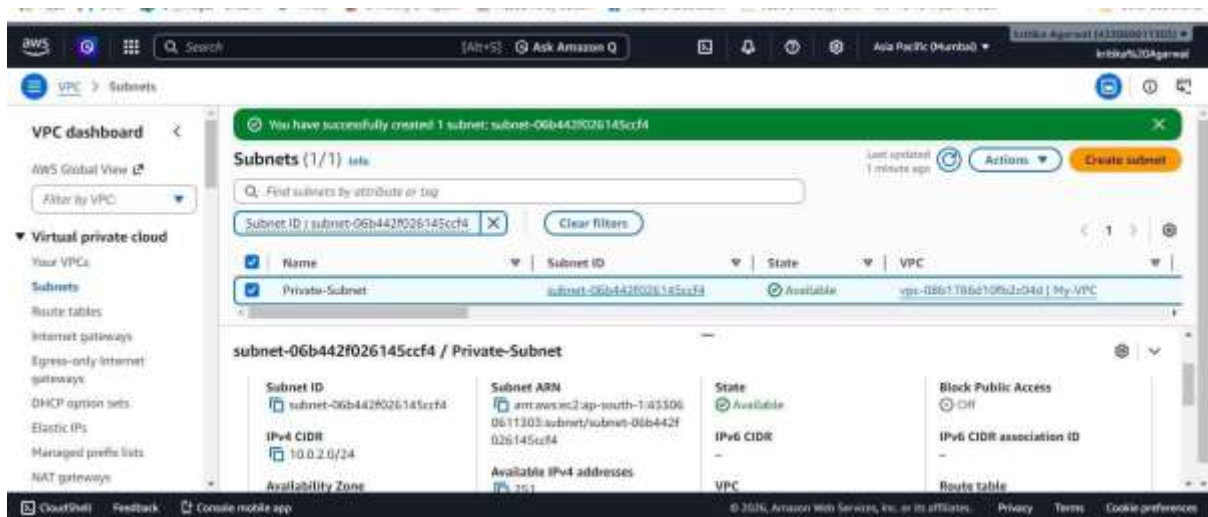


FIG-4: PRIVATE-SUBNET

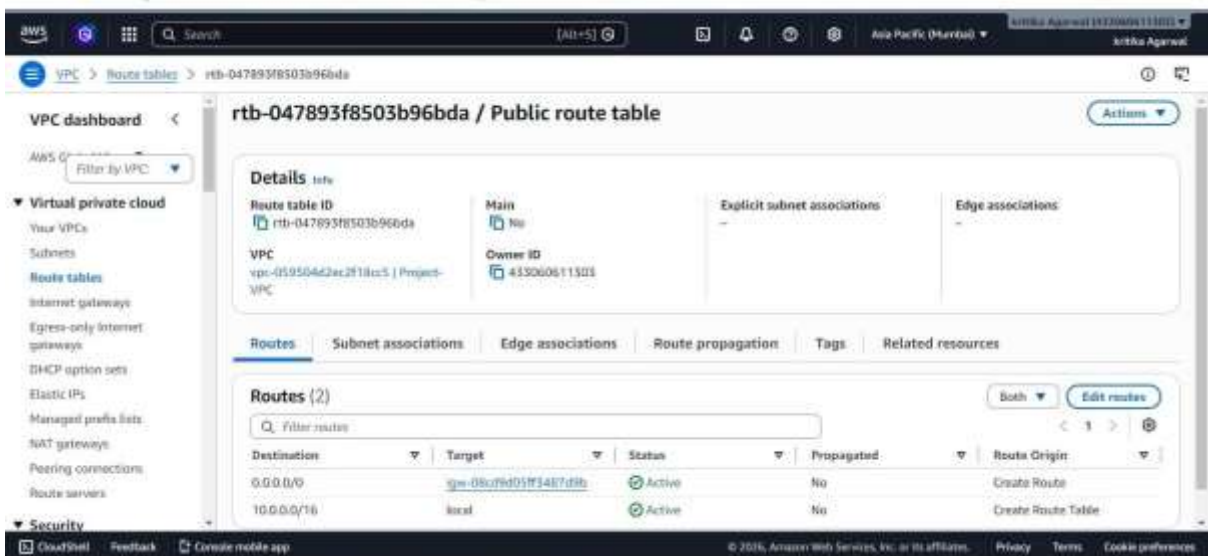


FIG-5: ROUTE TABLE

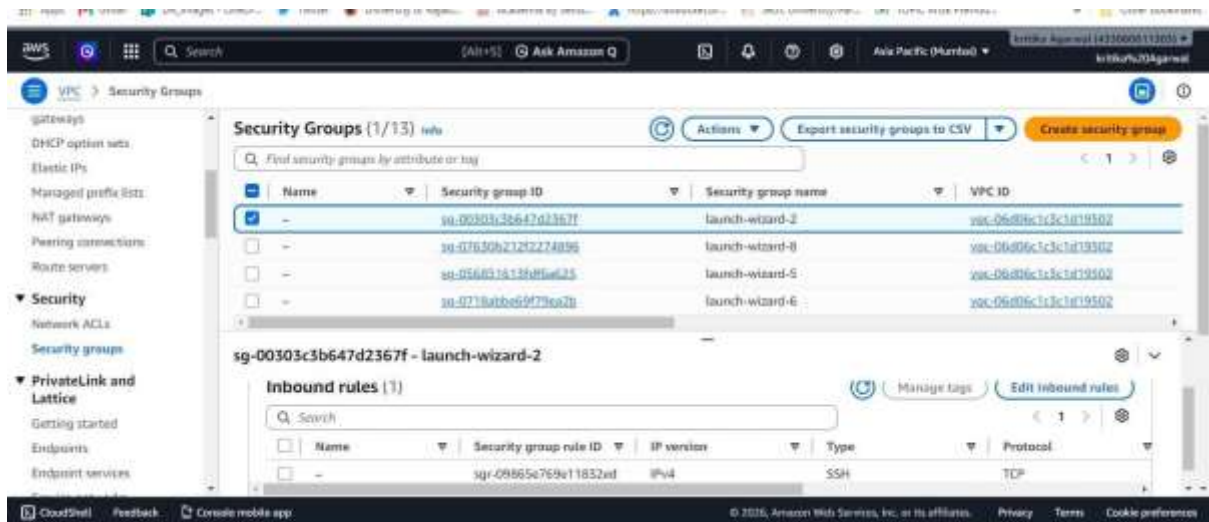


FIG-6: SG

Step 2: Launch Amazon EC2:

- Launch Amazon Linux EC2 instances.
- Install Apache or Nginx Web Server.
- Deploy the web application.
- Verify that the application is running successfully.

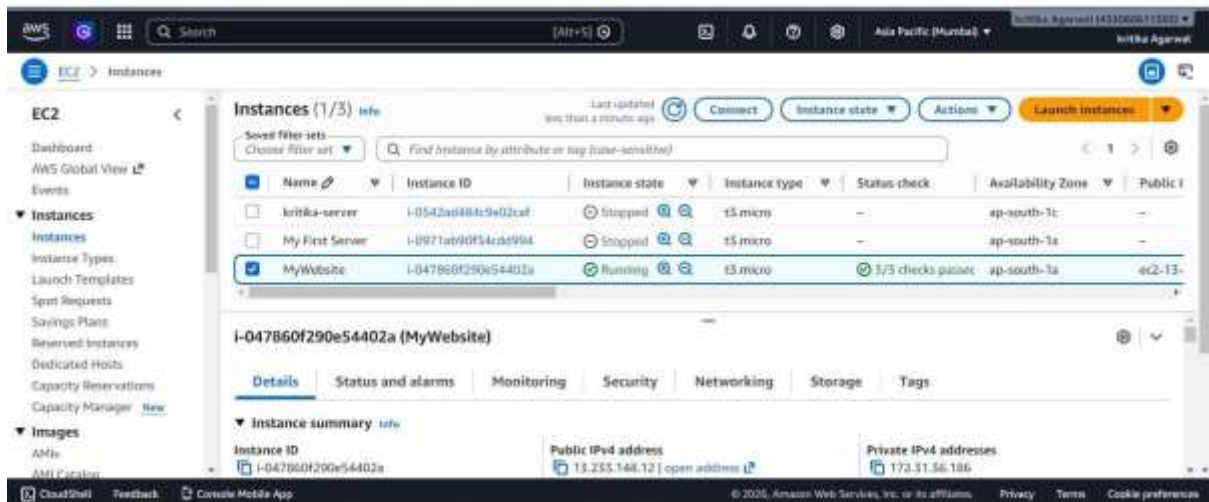


FIG: EC2 INSTANCE

Step 3: Configure Amazon RDS:

- Create a MySQL RDS database.
- Configure the RDS Security Group.
- Allow the EC2 Security Group to access the database on Port 3306.
- Connect the application with the RDS database.

```
root@RDS:~# mysql -h databasa01.c22icvge64th.ap-east-1.rds.amazonaws.com -u admin -p
Enter password:
1906 1945 (2000): Access denied for user 'admin'@'172.31.37.251' (using password: YES)
root@RDS:~# sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 5
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> ALTER USER 'root'@'localhost' IDENTIFIED BY 'RoshR12345';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> EXIT;
Bye
root@RDS:~# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 6
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

FIG: CREATE MYSQL

Step 4: Configure Amazon S3:

- Create an S3 bucket.
- Upload static files such as images, CSS, JavaScript, or backups.
- Configure the required permissions.

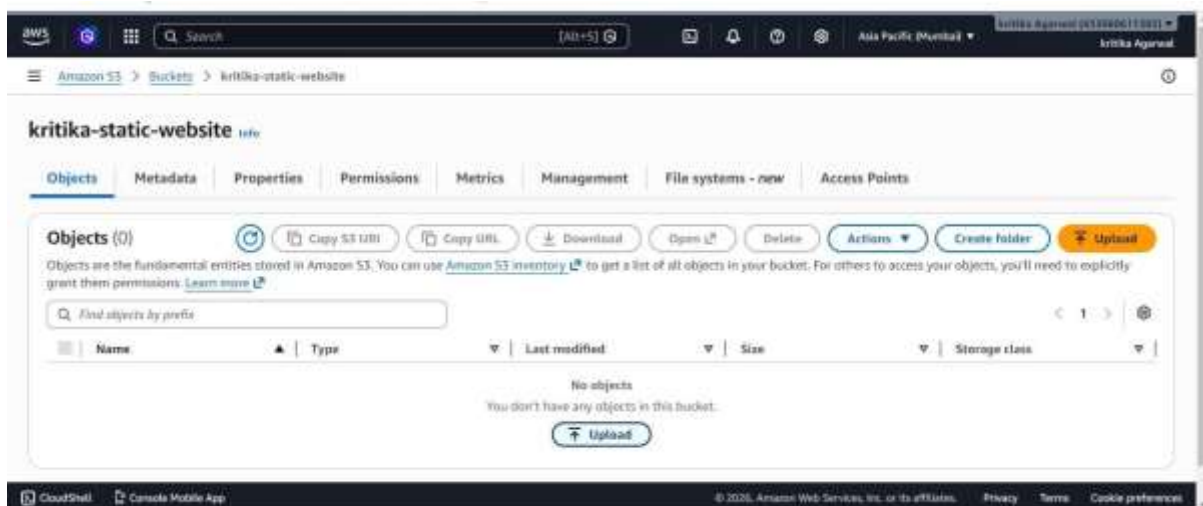


FIG: S3 BUCKET

Step 5: Configure Application Load Balancer:

- Create a Target Group.
- Register EC2 instances.
- Create an Application Load Balancer.
- Attach the Target Group.

- Verify that traffic is distributed across multiple EC2 instances.

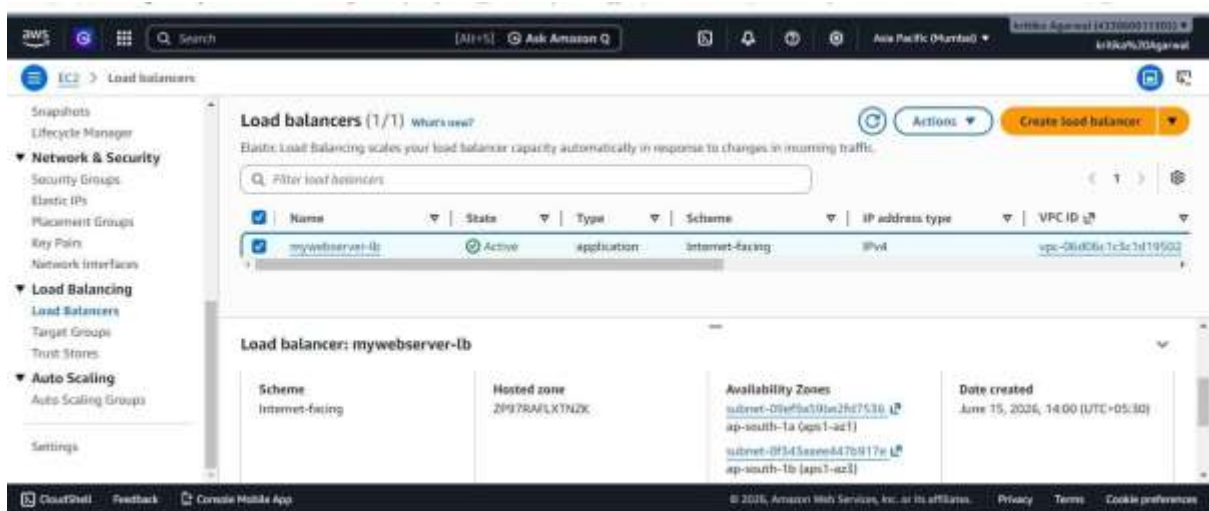


FIG: ALB

Step 6: Configure Auto Scaling Group:

1. Create a Launch Template.
2. Create an Auto Scaling Group.
3. Configure:
 - Minimum Capacity
 - Desired Capacity
 - Maximum Capacity
4. Configure the Target Tracking Scaling Policy based on CPU Utilization.

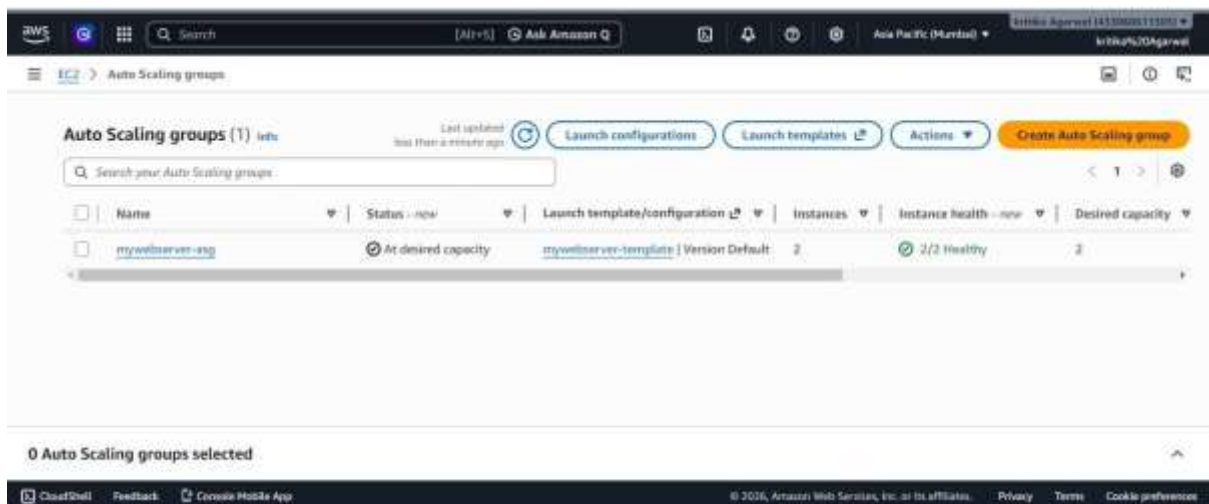


FIG:ASG

Step 7: Configure Route 53:

- Create a Hosted Zone.
- Create an Alias Record.
- Map the domain name to the Application Load Balancer.

Step 8: Configure CloudWatch Monitoring:

- Create CloudWatch Alarms.
- Monitor:
 - CPU Utilization
 - Network Traffic
 - Status Check
 - Application Performance

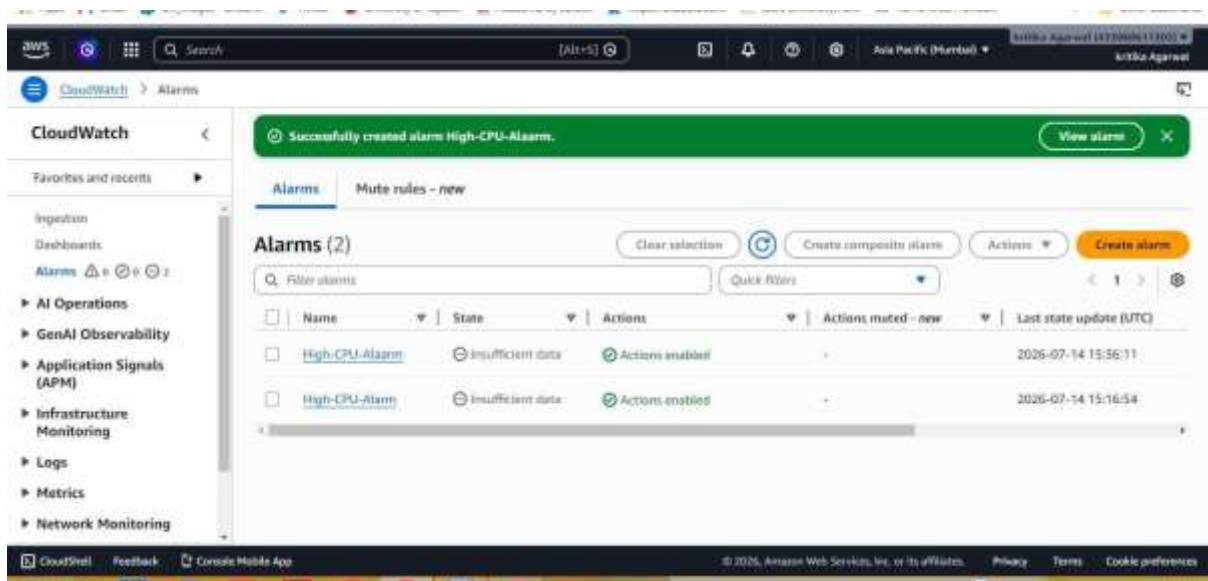


FIG: CLOUD WATCH

Step 9: Configure Amazon SNS:

- Create an SNS Topic.
- Subscribe an Email Address.
- Connect CloudWatch Alarms with SNS.
- Verify that notifications are received when alarm thresholds are exceeded.

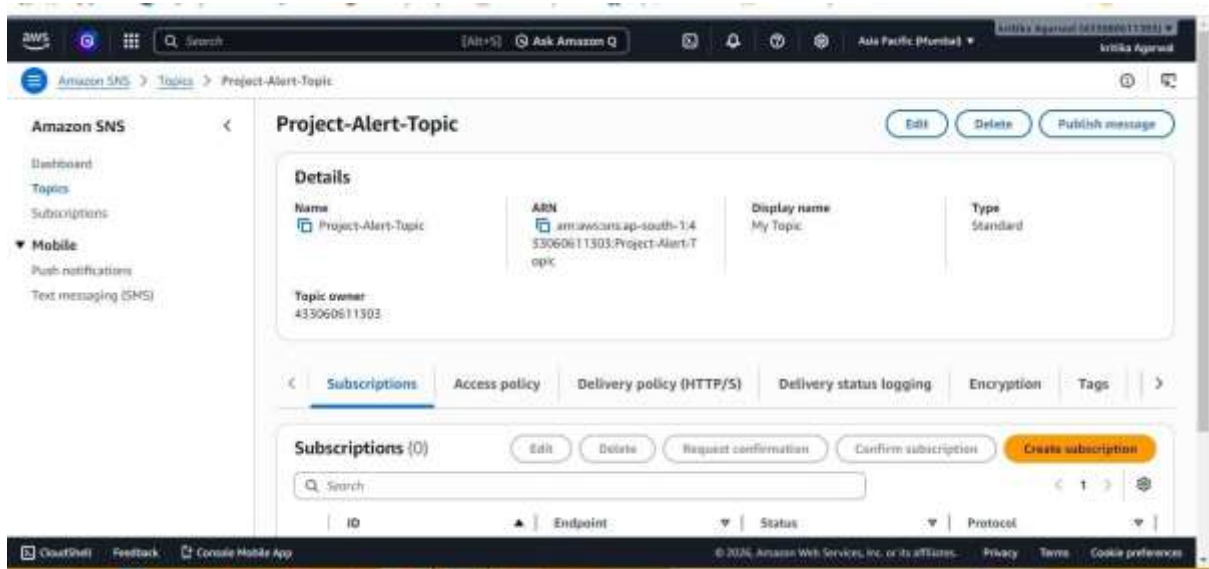


FIG: AMAZON SNS

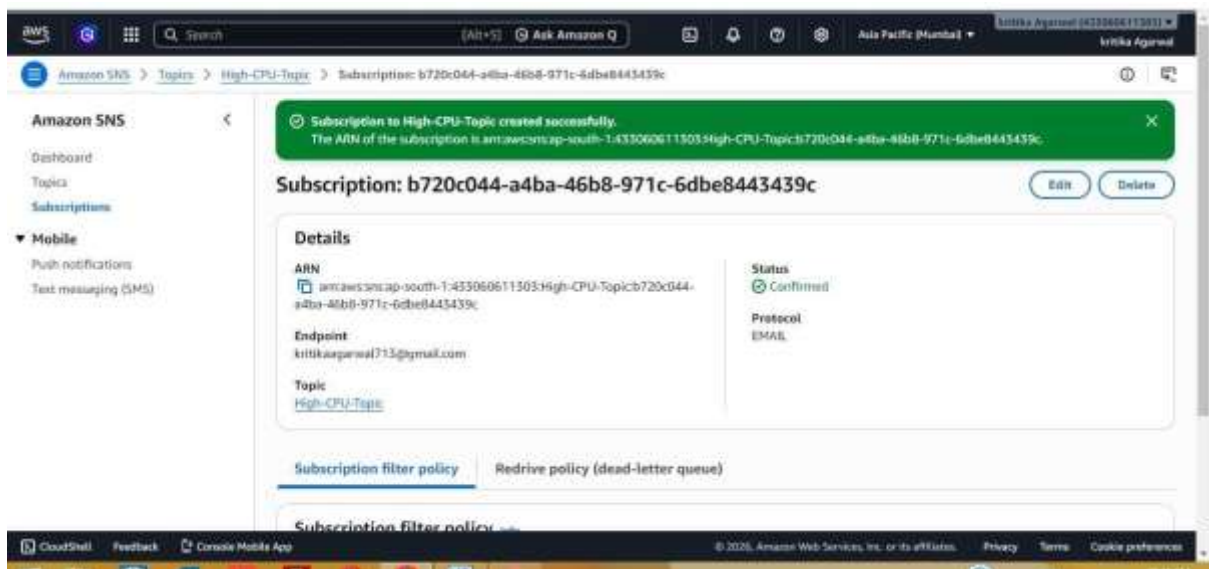


FIG: SUBSCRIPTION

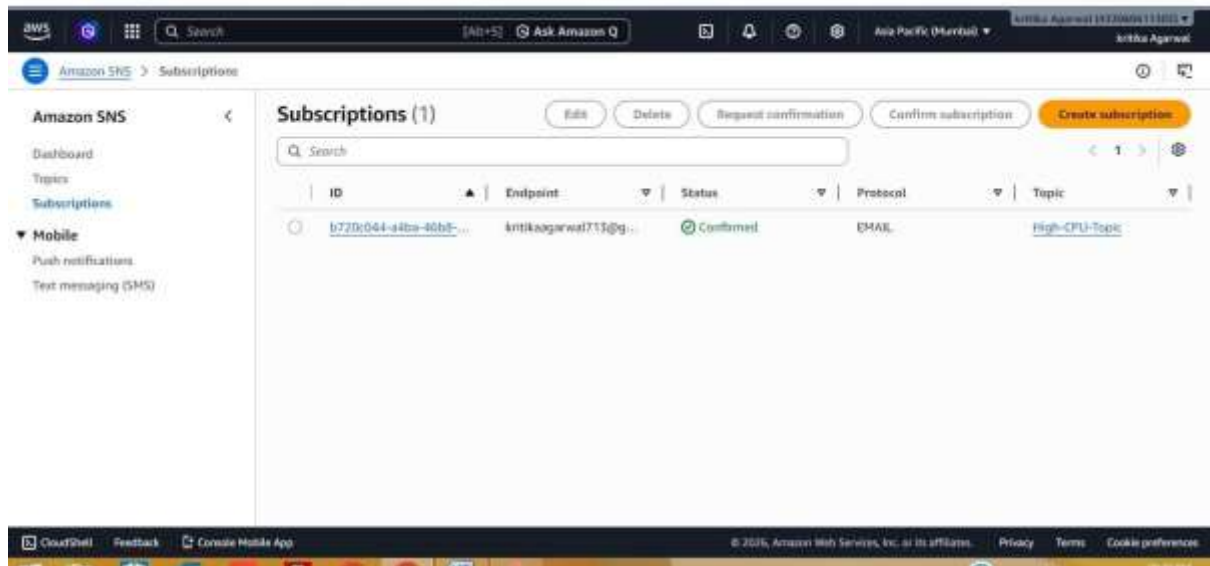


FIG:SNS

4. TESTING OUTCOME:

Step 10: Test the Application:

- Open the domain name in a web browser.
- Verify that the application loads successfully.
- Refresh multiple times to confirm Load Balancing.
- Terminate one EC2 instance.
- Verify that Auto Scaling launches a replacement instance.
- Trigger a CloudWatch Alarm and verify that an SNS email notification is received.

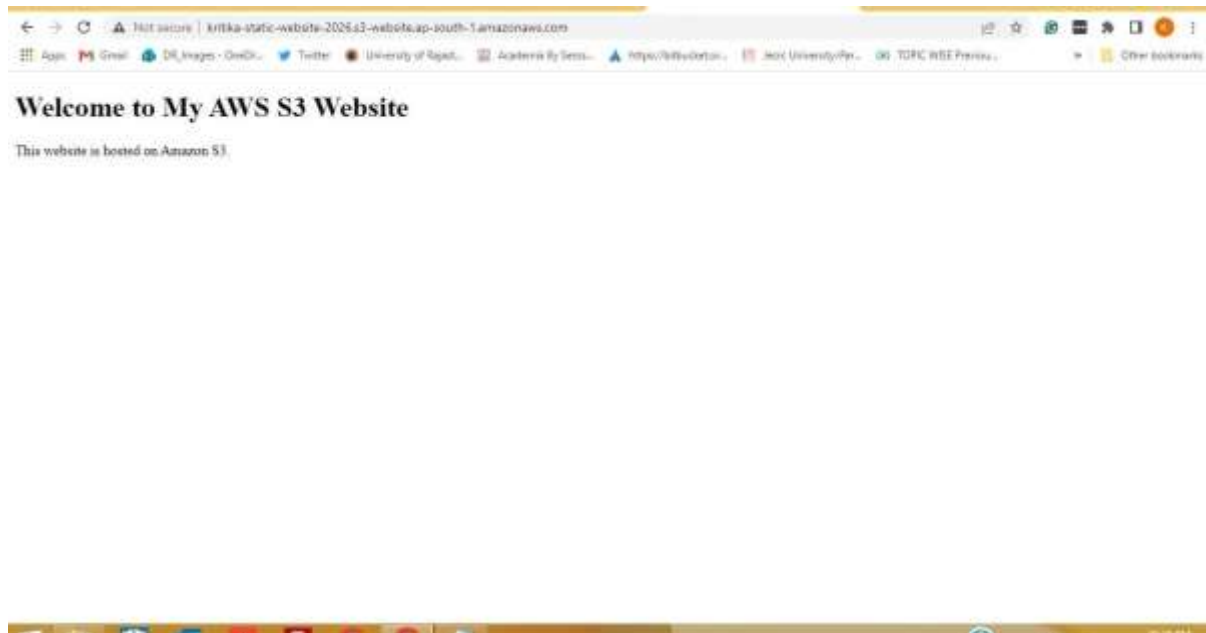


FIG:TESTING

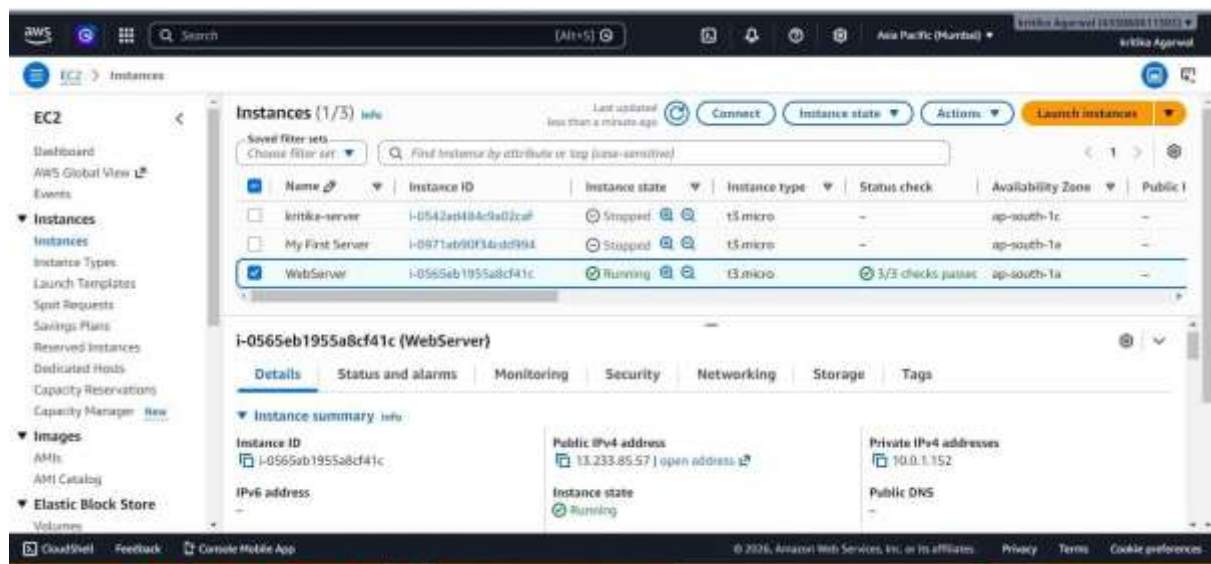


FIG:EC2 INSTANCE

5. CONCLUSION:

SUMMARY-

- Web application is deployed successfully.
- Load Balancer distributes incoming traffic.
- Auto Scaling launches additional instances during high traffic
- Amazon RDS stores application data securely.
- Amazon S3 stores static files.
- CloudWatch monitors application performance.
- SNS sends alert notifications.

LEARNING-

- How to design a three-tier cloud architecture.
- How to deploy applications on Amazon EC2.
- How to configure an Application Load Balancer.
- How to configure Auto Scaling.
- How to use Amazon RDS for database management.

THANK YOU
